GLA UNIVERSITY MATHURA

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12-B Status from UGC

BCA IVth Semester
Computer Communication and Networks
Course Code: BCAC0011

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Network
The Network Layer of the OSI Model is responsible for providing logical addressing, which routers use to select best path for routing packets. There are two types of packets used at this layer :
Data Packets –
The user data is transferred in the inter-network by these data packets. **Routed protocols** are those protocols which support such data traffic. Examples of routed protocols are IPv4, IPv6 and AppleTalk.

Route Update Packets –
The information about the networks connected to all the routers is updated to the neighbouring routers through route update packets. **Routing protocols** are the ones that are responsible for sending them. Examples of routing protocols are RIP(Routing Information Protocol), EIGRP(Enhanced Interior Gateway Routing Protocol) and OSPF(Open Shortest Path First).

ROUTING PROTOCOLS

- Routing protocols are the set of rules used by the routers to communicate between source & destination. They do not move the information source to destination only update the routing table.
- Each protocol has its own algorithm to choose the best path.

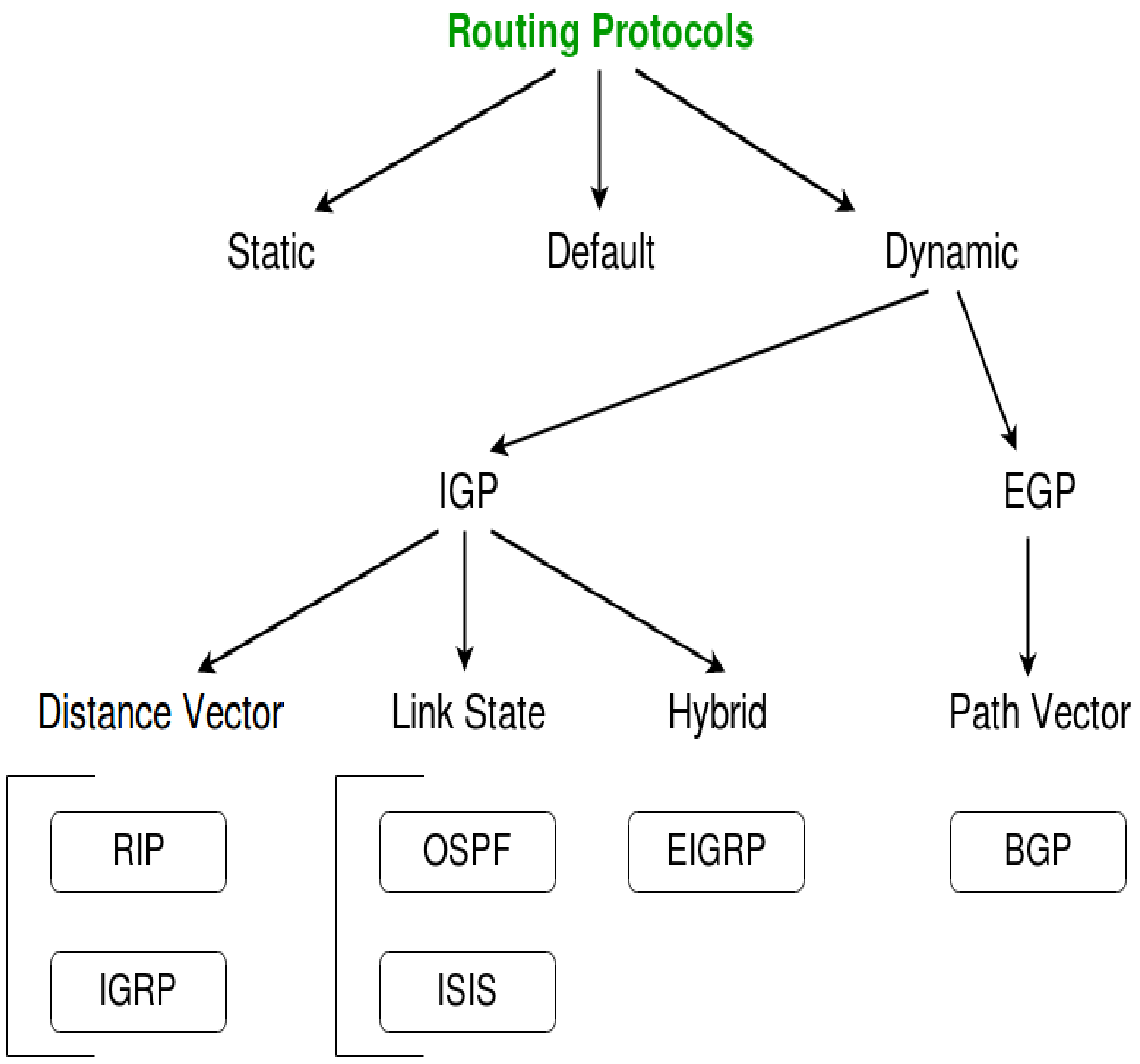
The metrics by routing protocols:-

- Number of network layer devices along with the path (hop count)
- Bandwidth
- Delay
- Load

TYPES OF ROUTING PROTOCOL

DYNAMIC ROUTING PROTOCOL

STATIC ROUTING PROTOCOL


Routing Protocols
Static
Default
Dynamic
IGP
EGP
Distance Vector
Link State
Hybrid
Path Vector
RIP
IGRP
OSPF
ISIS
EIGRP
BGP

Abbreviations –

IGP – Interior Gateway Protocol
EGP – Exterior Gateway Protocol
RIP – Routing Information Protocol
IGRP – Interior Gateway Routing Protocol
OSPF – Open Shortest Path First
ISIS – Intermediate System to Intermediate System
EIGRP – Enhanced Interior Gateway Routing Protocol
BGP – Border Gateway Protocol

STATIC ROUTING PROTOCOLS

Static routing ,when an administrator manually assigns the path from source to destination network.
This is feasible in small networks, but not in large networks.
Advantages: No overhead on router CPU.
No bandwidth usage between links.
Security (only administrator add routes.)

Disadvantages: All link will be down on a link failure.
Not practical on large networks.
Administrator must update all routes.

DYNAMIC ROUTING PROTOCOLS

Dynamic routing is the process in which routing tables are automatically updates by routing table of each neighbor.

– Dynamically discover & maintains routes.

– Calculate routes

Advantages – Less work in maintaining the configuration when adding & deleting networks.

– Protocols automatically react to the topology changes.

– Configuration is less-prone.

Disadvantages – Routers resource are used.

– More administrator knowledge is required for configuration

DEFAULT ROUTING PROTOCOLS

Default Route is the network route used by a router when there is no other known route exists for a given IP datagram’s destination address.
All the IP datagrams with unknown destination address are sent to the default route.

- **Advantages**:- No overhead on router CPU.

– No bandwidth usage between links.
– Security (only administrator add routes.)

- **Disadvantages**:- All link will be down on a link failure.

– Not practical on large networks.
– Administrator must update all routes.

Routing Concepts:

There are some concepts which are using in routing. That are

1) Least-cost routing:

Router chooses the best possible path on the basis of **least cost routing.** The decision of the least cost routing is based on efficiency, i.e. which of the available path is cheapest or shortest. ***The term shortest means in some case, the route requiring the smallest number of intermediate nodes or in other case it should be cheapest, most reliable, and most secure.***

Routing is classified in two types

a) Non adaptive Routing:

In some routing protocol, once a pathway to a destination is selected, the router sends all packets for that destination along that route only.

b) Adaptive Routing:

In the adaptive routing technique, router may select a new route for each packet (even packet belonging to the same transmission) on the basis of condition and topology.

Packet lifetime:

Once a router has decided a route, it passes the packet to the next router on that route and forgets about it. The next router however may choose the same route or may be different and relay the packet to the next router in that direction.

This process may cause of loops and bouncing when a router updates its routing table, then relays a packet based on new route.

A field is added to the packet called packet life time, or time to live (TTL). As it is generated, each packet is marked with a lifetime; usually the number of hops that are allowed before a packet is considered lost and accordingly destroyed.

1) Distance vector routing:

In distance vector routing, each router timely shares its knowledge about the entire network with its neighbor. The three keys to understanding how this algorithm works are as follows:

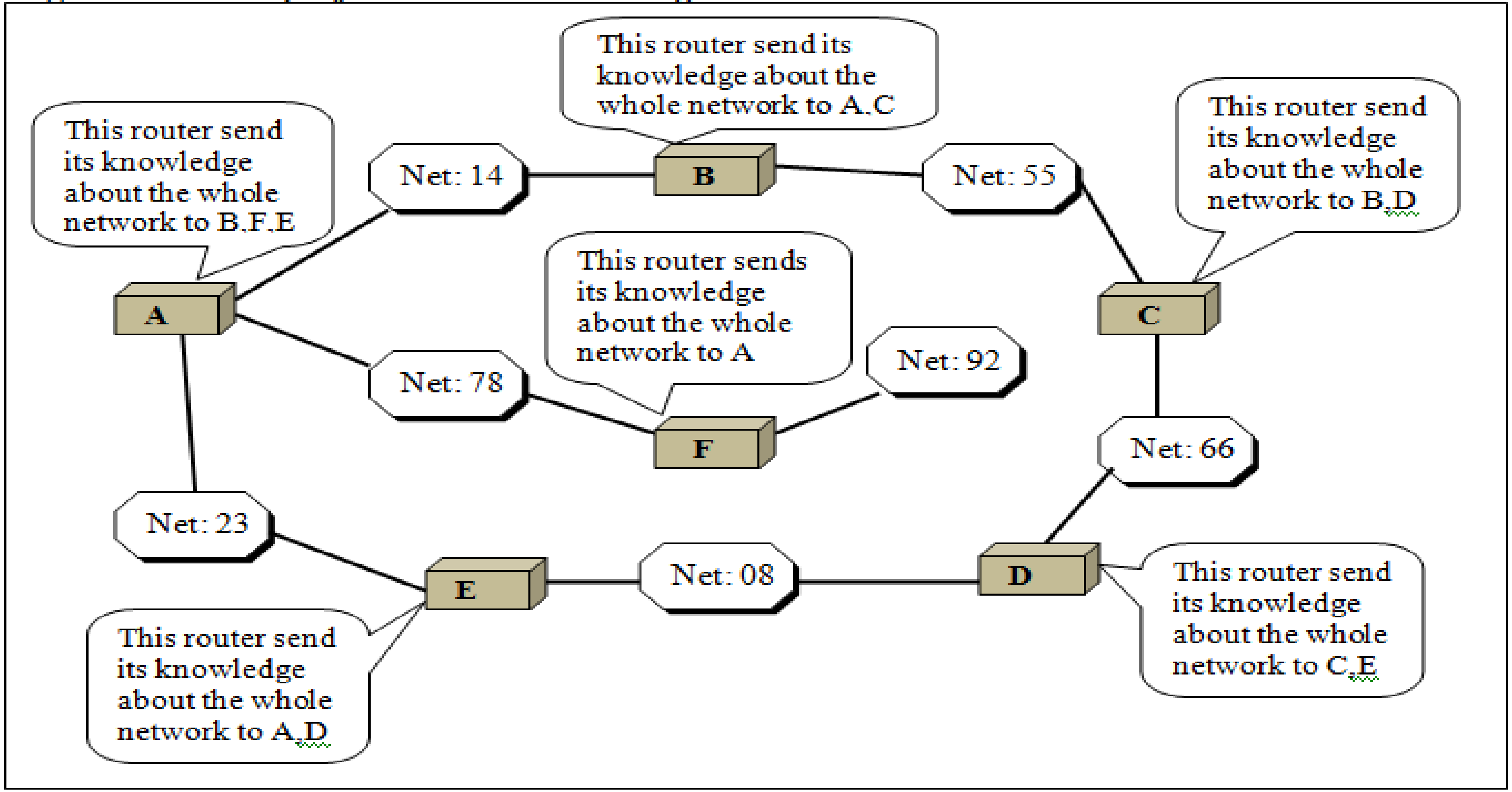
Knowledge about the whole network: Each router has whole knowledge about the entire network .

Share information only to neighbor: each router timely shares its whole knowledge about the network only to those router to which it has direct link. This information is received and kept by each neighbor router and used to update that own information about the network.

Information sharing at regular intervals: Each router sends its information about the whole network to its neighbor at regular interval. This sharing occur whether or not the network has changed since the last time information has exchanged.

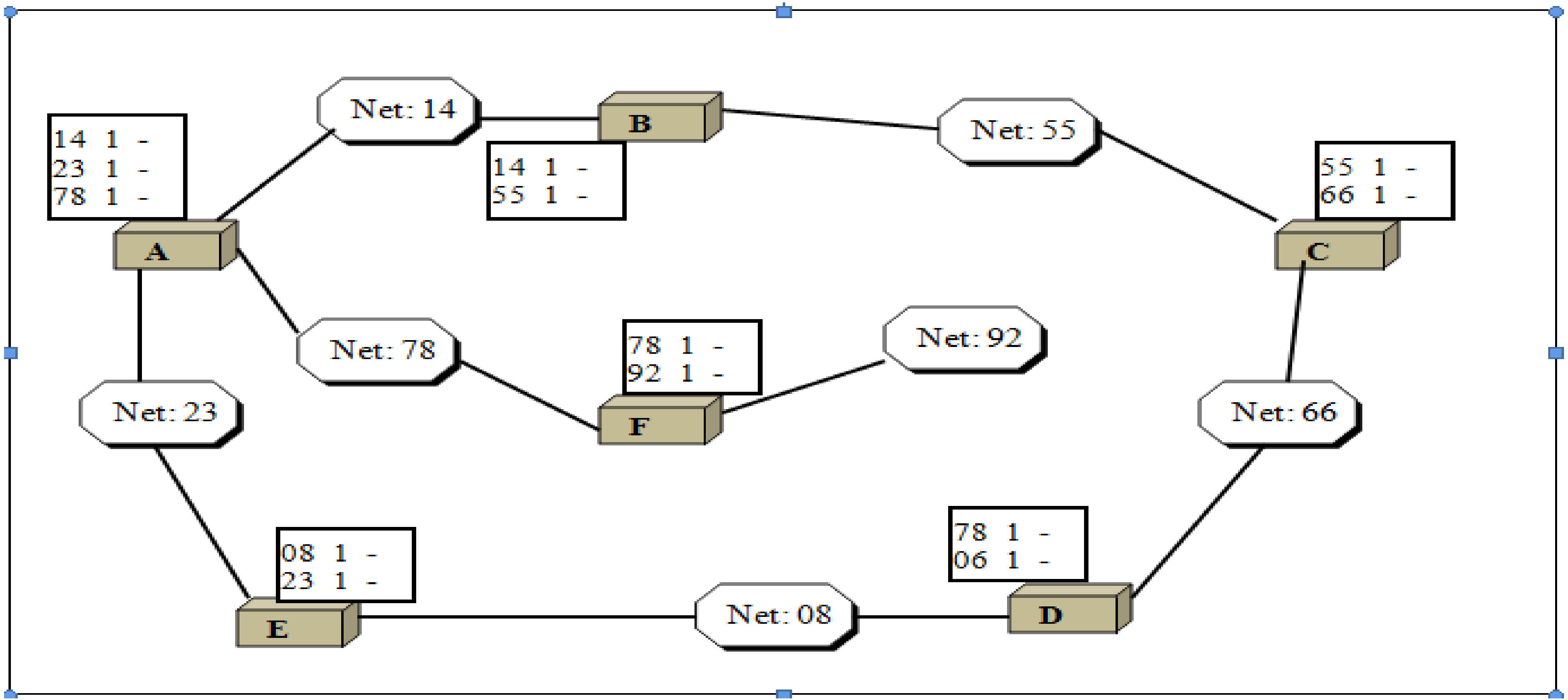
Distance vector routing simplifies the routing process by assuming a cost of one unit for every link.

Figure: ***The concept of distance vector routing***

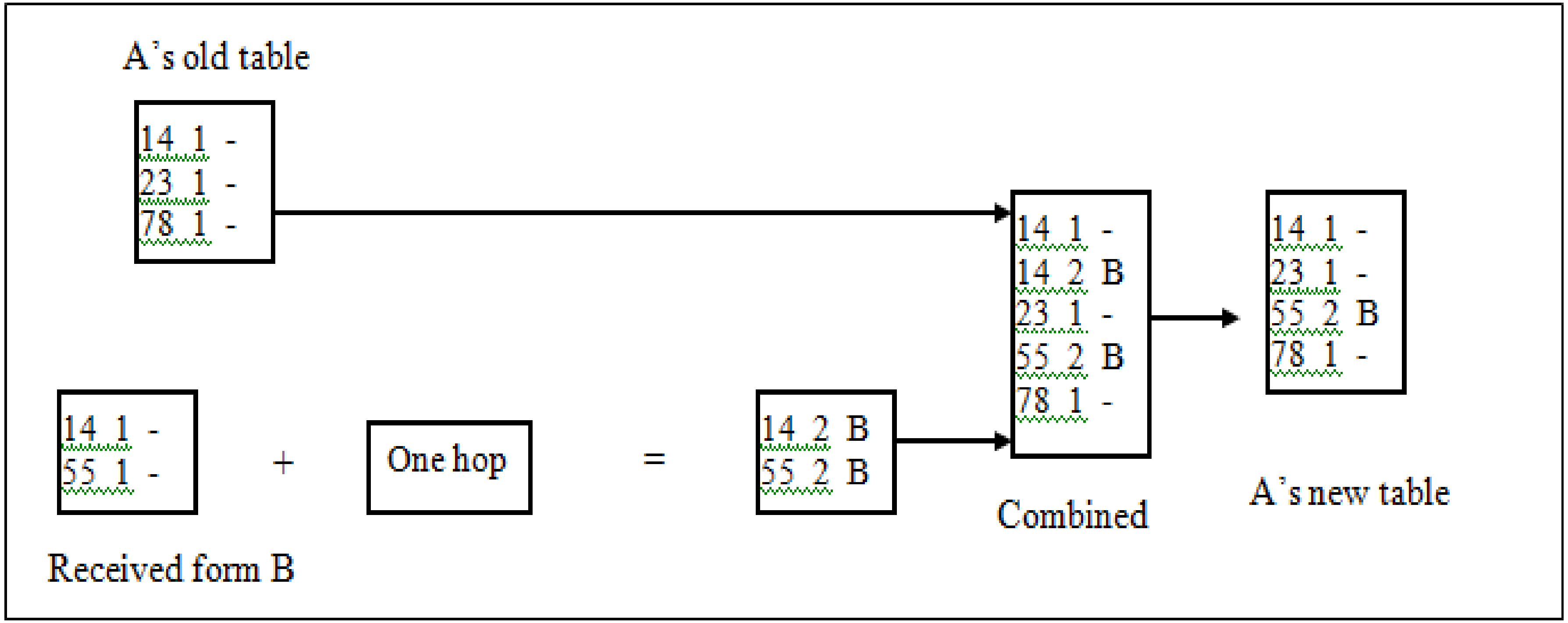


Routing Table:

Each router maintains a table called routing table, which has the information as the network ID, the cost, and the ID of the next router (next hop). The network ID is the identification number of network through which the router is connected. The cost is the number of hops between the routers from source to destination. And the next router is the router to which a packet must be delivered on its way to particular destination. The table tells a router that it costs x to reach network Y via router Z.

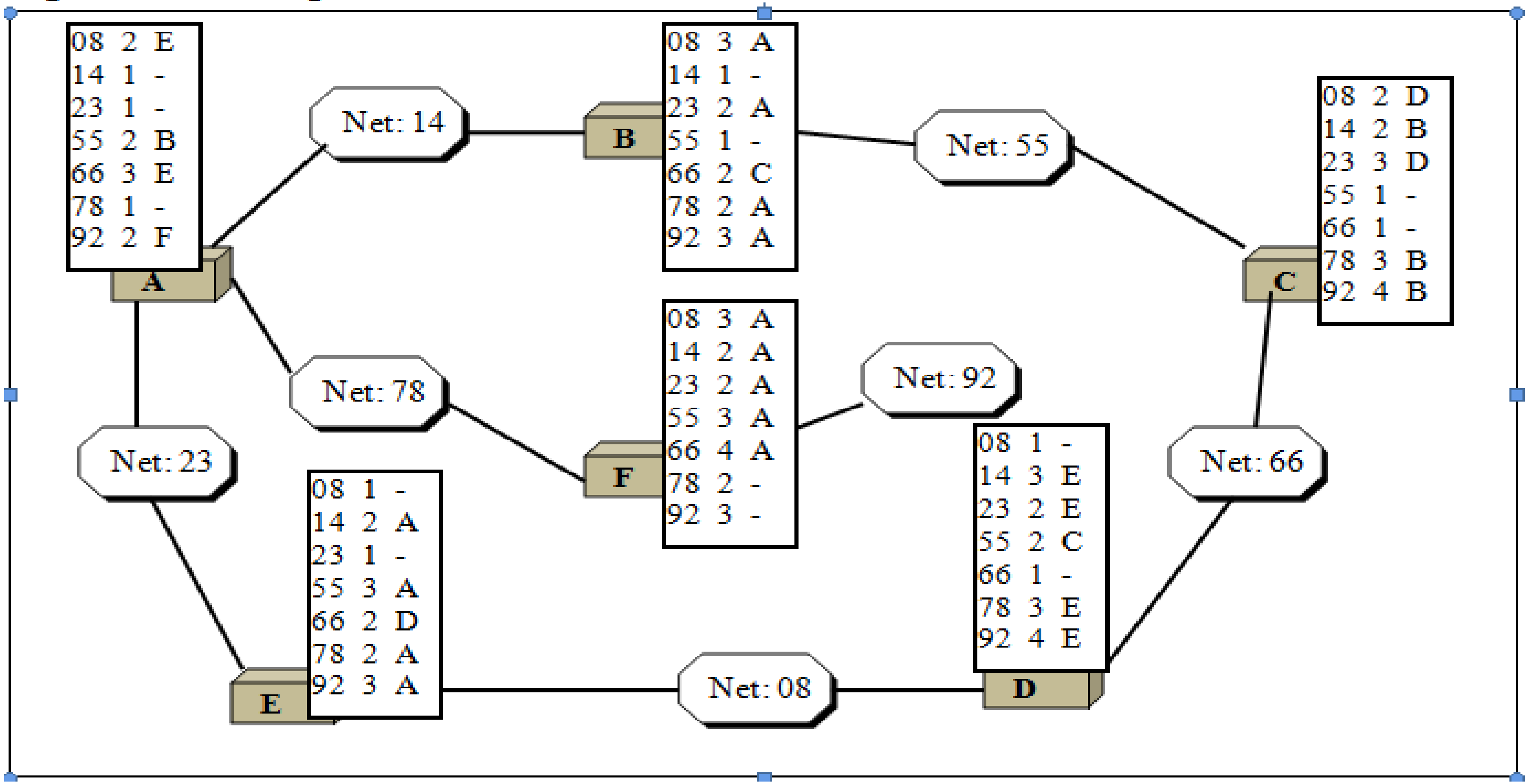


Updating routing table:
When A receives a routing table from B, it uses the information to update its own table. As



The updated table may contain duplicate data for some network destinations. Router finds and removes any duplication and keeps which ever version shows the lowest cost.

Figure: *Final routing tables*



Updating Algorithm:

The updating algorithm requires that the router first add one hop to the hop count field for each advertised route. The router should then apply the following rules to each receiving route.

1. If the receiving destination is not in the routing table, the router should add the received information to the table.
2. If the received destination is in the routing table :

   If the next hop field is same, the router should replace the entry in the table with the received one. Note that even if the received hop count is larger. The received entry should replace the entry in the table because the new information invalidates the old.

If the next hop filed is not the same

i) If the received hop count is smaller than the one in the table, the router should replace the entry in the table with the new one.

ii) If the received hop count is not smaller (same or large) the router should do nothing.

Message from C

Net2	4
Net3	8
Net 6	4
Net 8	3
Net 9	5

Message from C after increment

Net2	5
Net3	9
Net 6	5
Net 8	4
Net 9	6

Rules:
Net2: Replace (Rule 2.a)
Net 3: Add (Rule 1)
Net 6: Replace(Rule 2.b.i)
Net 8: No change(Rule 2.b.ii)
Net 9: No change(Rule 2.b.ii)

Old routing table

Net1	7	A
Net 2	2	C
Net 6	8	F
Net 8	4	E
Net 9	4	F

Updating table

Net1	7	A
Net2	5	C
Net3	9	C
Net 6	5	C
Net 8	4	E
Net 9	4	F

Note: that there is no news about Net1 in the receiving message, so none of the rules apply to this entry

2) Link state routing:

In **link state routing,** each router shares its knowledge of its neighborhood with every other router in the internetwork. This algorithm works as follow

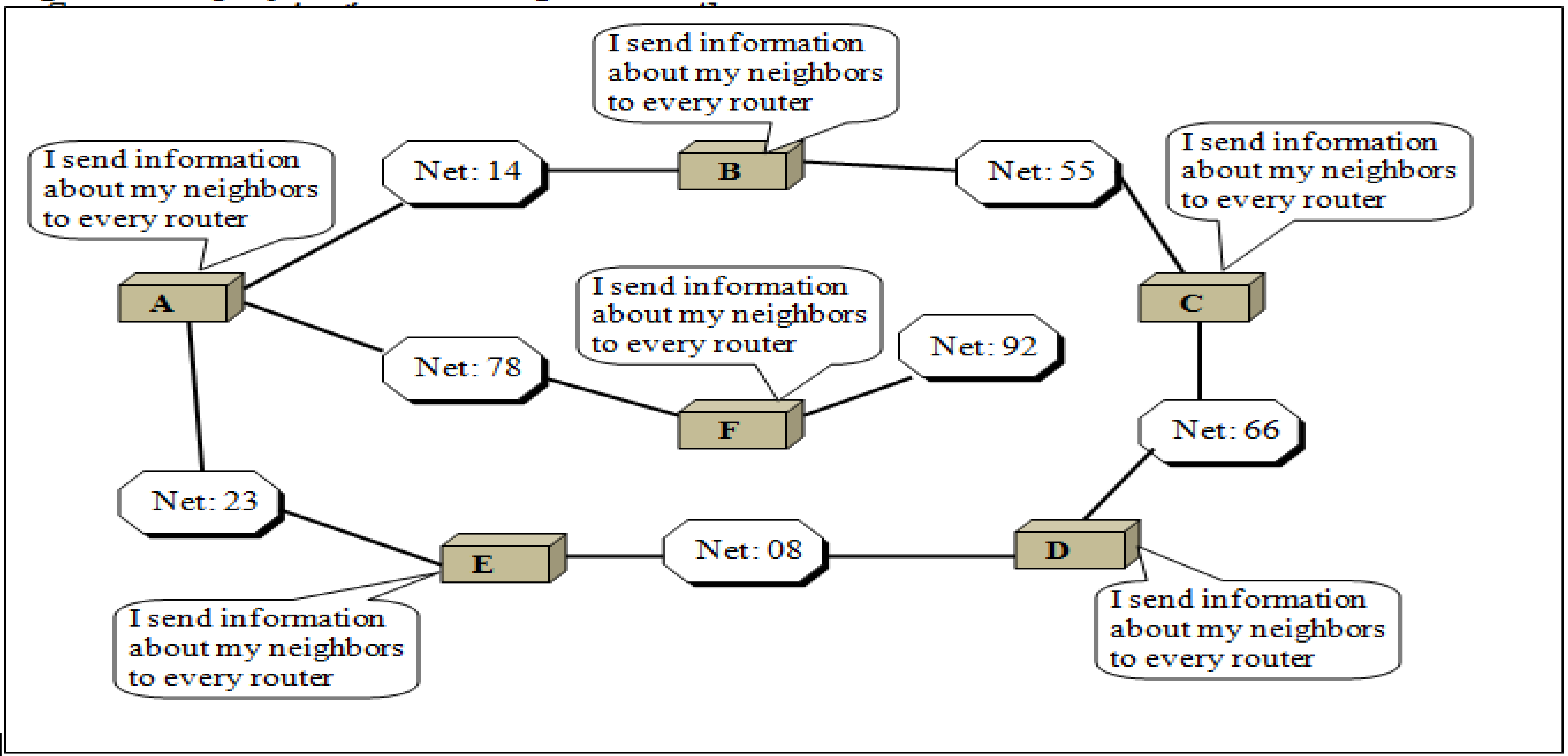
i) Knowledge about the neighborhood: instead of sending its entire routing table, a router sends information about its neighborhood only.

ii) To all router: Each router sends this information to every other router on the internetwork, not just to its neighbors only. This process is called **flooding.**

Flooding means that a router sends its information to all of its neighbors, each neighbor sends the packet to its entire neighbor and so on and finally every router receives a copy of the same information.

iii) Information sharing: each router sends out information about the neighbors when there is a change.

Figure: ***concept of link state routing***



In link state vector routing cost is based upon different factor such as security levels, traffic or state of the link. The cost from router A to network 14 might be different from the cost from A to 23.

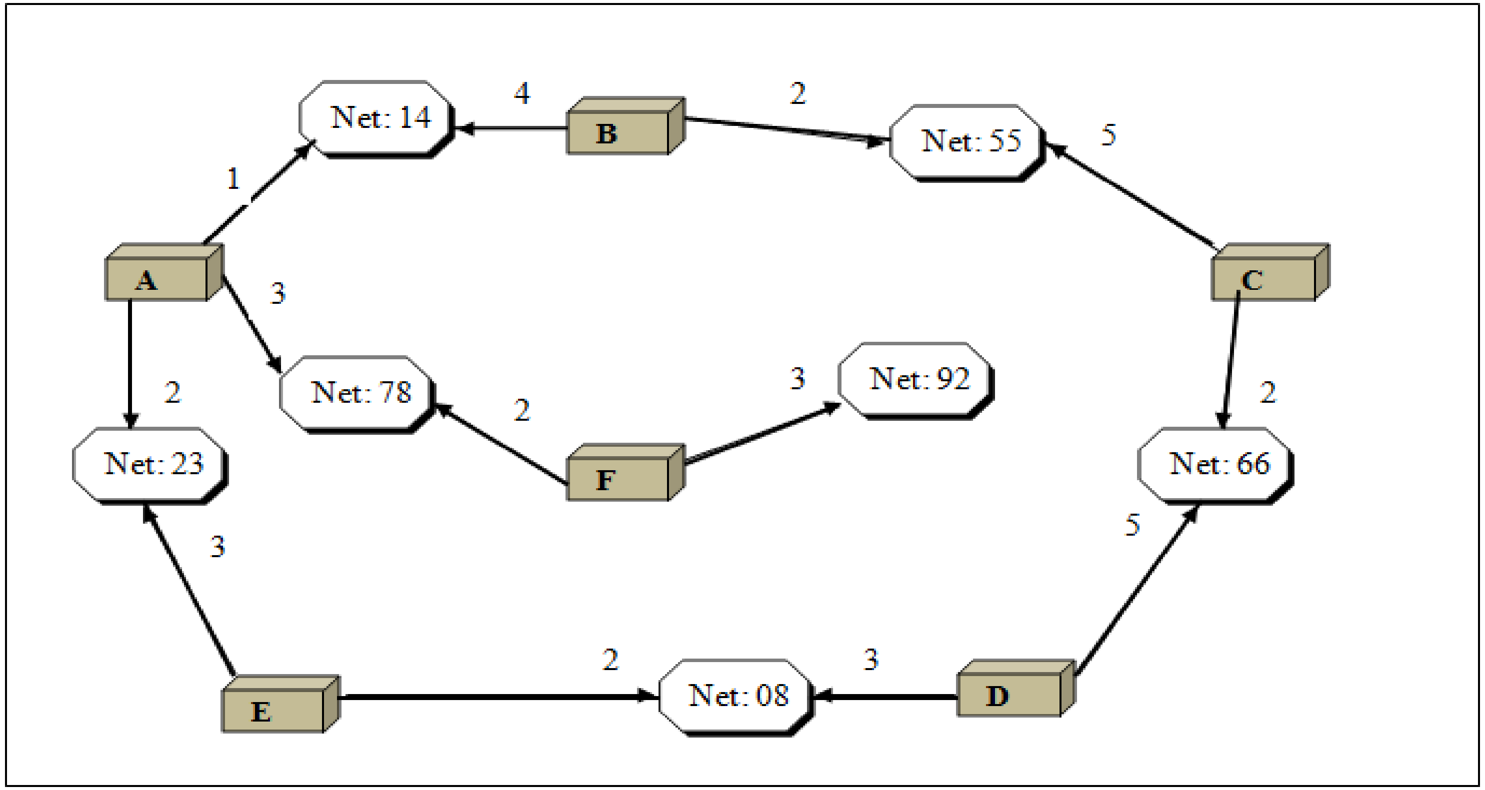
Link state packet:

When a router has the network with information about its neighborhood. It is said to be advertising. The basis of this advertising is a short packet called a link state packet (LSP). An LSP contains four fields: the ID of the advertiser, the ID of the network, the cost and the ID of the neighbor router.

In determining a route, the cost of a hop is applied to each packet when it leaves a router and enters a network. Two factors govern how cost is applied to packet to determining a route.

•Cost is applied only by routers and not by any other stations on a network.

•Cost is applied as a packet leaves the router than as it enters.



Link state packet:

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Figure: *link state packet*

Advertising	Network	Cost	Neighbor
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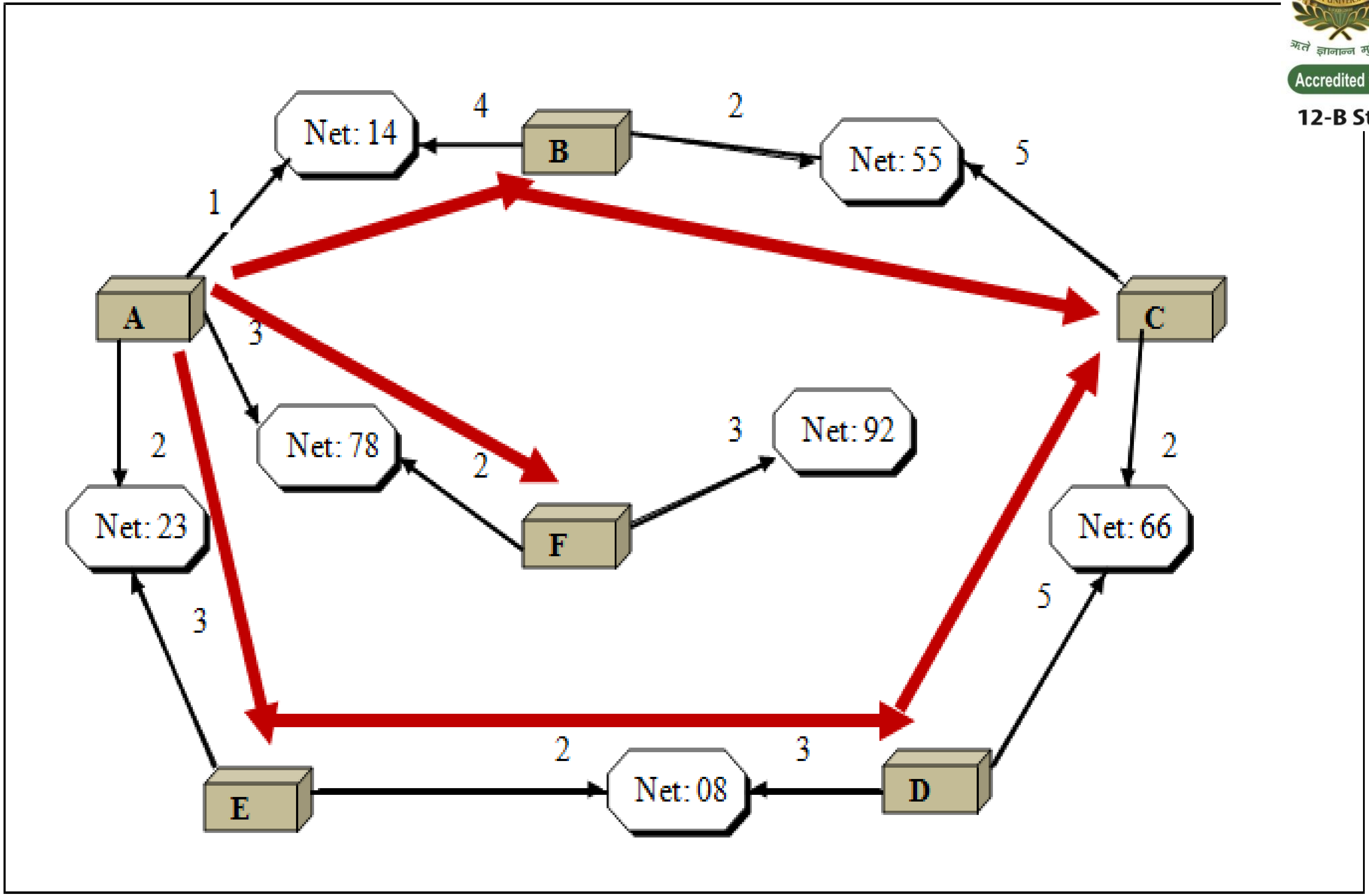
Figure: *Flooding of A's LSP*



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Link state database:
Every router receives every LSP and puts the information in to a link state database.

Figure: ***Link state database***

Advertiser	Network	Cost	Neighbor
A	14	1	B
A	78	3	F
A	23	2	E
B	14	4	A
B	55	2	C
C	55	5	B
C	66	2	D
D	66	5	C
D	08	3	E
E	23	3	A
E	08	2	D
F	78	2	A
F	92	3	■

Because every router receives the same LSPs, every router builds the same database. It stores this database on its disk and use it to calculate its routing table. If a router is added or deleted from the system, the whole database must be shared for fast updating.

Dijkstra Algorithm:

To calculate its routing table, each router applies an algorithm called the dijkstra algorithm to link state database. The dijkstra algorithm calculates the **shortest path** between two node on a network using a graph.

a) Shortest path tree:

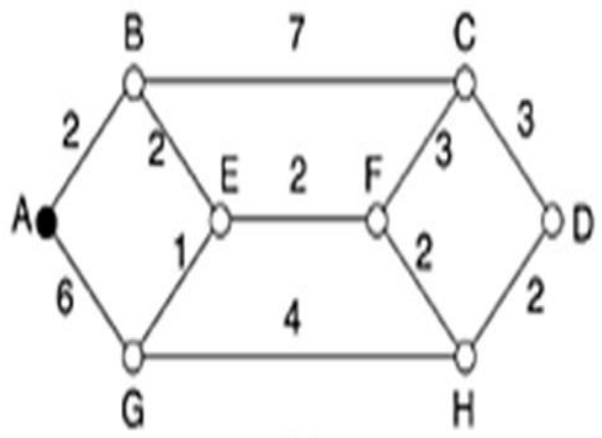
The dijkstra algorithm follows four steps to discover what is called the **shortest path tree** for each router.

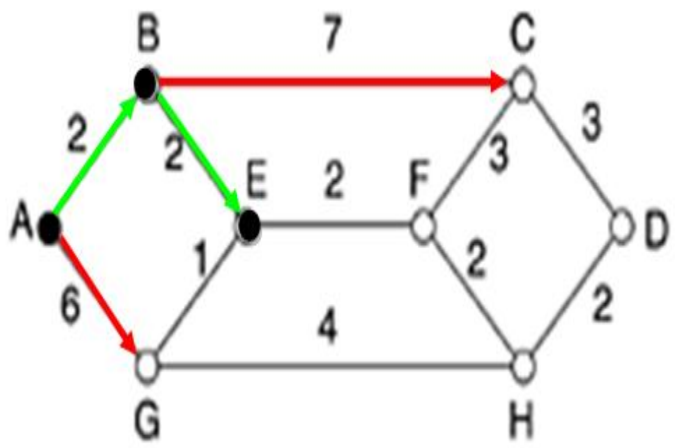
1) The algorithm begins to build the tree by identifying its root. The root of each router's tree is the router itself.

2) In next step all possible paths are found to reach the next node from the root node towards the destination. And identifies the shortest path and reach one step ahead.

3) Now find the all possible path for next node from that root and check the same for previous node but make sure that a complete cycle should not make. If from previous node the path for next node is shortest then this path is selected and reaches one step ahead.

4) Repeat step 3 until reach to final destination.


B
7
C
2
2
3
3
A
E
2
F
D
1
2
6
4
2
G
H
(a)


B
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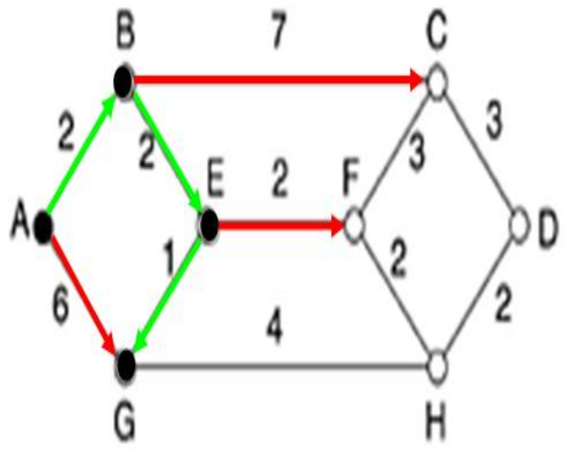
b.


B
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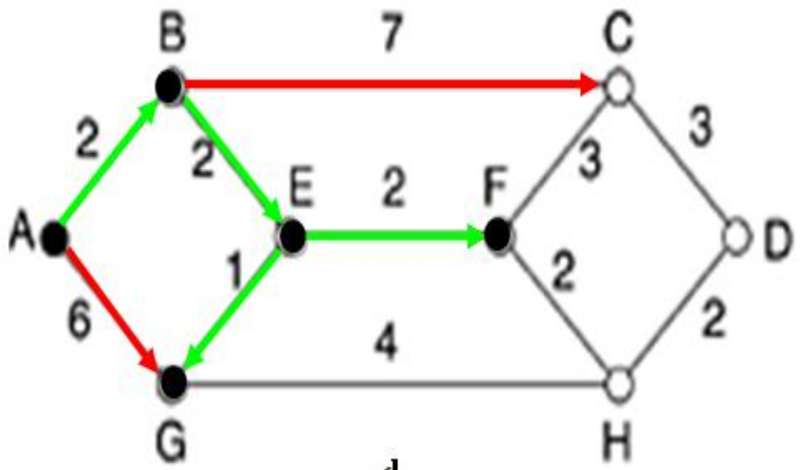
B
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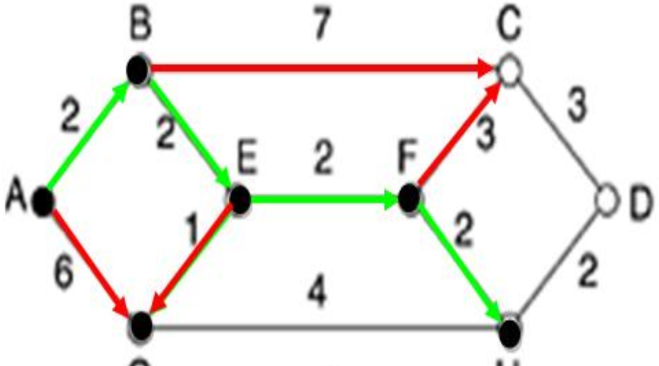
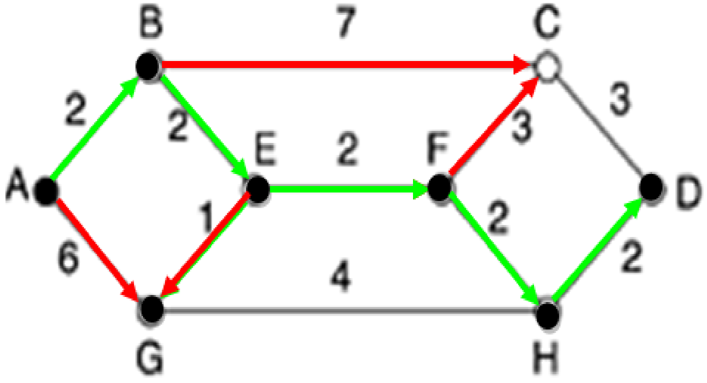


e



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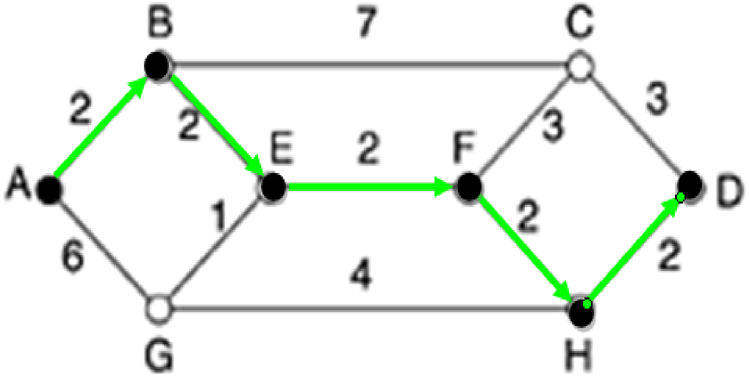

g.

Final route from source to destination